MILITARY SPECIFICATION

FIRE CONTROL SYSTEM FOR GUN, 155MM, SELF-PROPELLED, H53 AND HOWITZER, 8-INCH, SELF-PROPELLED, M55

1. SCOPE

1.1 Scope. - This specification covers the fire control system of the gun, 155mm, self-propelled, M53 and the howitzer, 8-inch, self-propelled, M55.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS
Military

MIL-G-10644	Gun, 155mm, Self-Repelled, T97 (W/E).
MIL-H-11964	Howitzer, 8-Inch, Self-Propelled, T108 (W/B).

ENGINEERING PARTS LISTS
Ordnance Corps

7393840	Gun, 155mm, Self-Propelled, M53.
7393841	Howitzer, 8-Inch, Self-Propelled, M55.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design. - The fire control systems shall conform to the drawings specified on Engineering Parts Lists 7393840 and 7494841.

3.2 Environmental requirements. - The fire control system shall be operable at minus 40° ± 2°F., minus 25° ± 2°F., plus 70 ± 7°F ., and 125° ± 2°., when

3.2.1 Storage. - The fire control system shall be operable at the temperatures specified in 3.2 after storage for 24 hours at minus 80°F and plus 160°F.

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3.3 Performance requirements.

3.3.1 Gun stops. - The gun shall engage stops at elevations of minus 89 to minus 71 mils anti plus 1157 to plus 1175 mils when tested as specified in 4.3.

3.3.2 Alinement of panoramic telescope. - When tested as specified in 4.4.1, the vertical crosshair of the telescope reticle shall not deviate from the target by more than 0.75 mil.

3.3.3 Levelness of the azimuth table, - The level bubbles of the cross leveling adjustments shall remain centered within plus or minus one-half vial graduation when tested as specified in 4.4.2. One graduation equals 0.1 inch.

3.3.4 Horizontal deflection of panoramic telescope. - When tested as specified in 4.4.3, the difference in reading after pressures have been applied to the eyepiece in opposite directions, shall not exceed 0.4 mil for zero cant and zero elevation and 2.5 mils for zero cant and 1100 mils elevation. The above tolerances shall apply at any position of the azimuth table.

3.3.5 Accuracy of elevation counter. - When tested as specified in 4.4.4, true weapon elevation, excluding backlash, shall vary from the elevation counter setting by not more than 0.5 mil at zero counter setting and by not more than 1.0 mil at all other counter settings.

3.3.5.1 Backlash.- When tested as specified in 4.4.4.1, backlash shall be not more than 1.0 mil at any elevation counter reading from minus 71 mils to plus 1157 mils.

3.3.5 Boresight check if panoramic telescope mount. - The line of sight of the panoramic telescope shall be within ± 3.0 mils of the target aiming point, when tested as specified in 4.4.5.

3.3.7 Direct sighting telescope mount. - When tested as specified in 4.5.1, the line of sight shall not deviate from a plumb line by more than 1.0 mil.

3.3.8 Boresight check of direct fire telescope. - When tested as specified in 4.5.2, the line of sight of the telescope shall lie on its respective target center within 5.0 mils.

3.3.9 Boresight retention. - When tested as specified in 4.6, the panoramic telescope line of sight shall have maintained its original setting with respect to the weapon tube in azimuth within ± 0.75 mil and its original setting of elevation counter with respect to the quadrant pads of the breech within ± 0.5 roil. The direct sighting telescope mount shall have maintained its original setting with respect to the tube within ± 0.5 mil in azimuth and ± 0.5 mil in elevation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Tests.

4.1.1 Environmental requirements. - To determine conformance to 3.2, the fire control system shall be tested in accordance with the procedures specified in 4.3 through 4.5. Tests shall be performed at a Government proving ground.

4.1.2 Acceptance. - To determine conformance to 3.3, the fire control system shall be tested in accordance with the procedures specified in 4.3 through 4.6. Tests shall be performed at existing air temperatures.

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4.2 Equipment.

(a) Mechanical jacks or leveling platform for leveling the vehicle.
(b) Plumb line, 1/16 inch in diameter, shall be suspended 15 feet from the muzzle of the weapon. The line shall be shielded from wind currents and the plumb bob shall be immersed in a container of oil to act as a dampener.
(c) A boresight retention and checking target shall. be constructed of non-warping and non-shrinking material designed as shown in figure 1. The target shall be placed at a distance of 50 feet from the center of the weapon trunnions in a plane perpendictir to the weapon bore at zero gun elevation. the reference line on the target shall be plumb. The height of the target above the ground shall be such that, at zero gun elevation the center-line of the gun bore shall fall on the weapon aiming point within ± 1/8 inch.
(d) An approved bore scope (see 6.2) shall be used. This shall be a telescope with cross hair reticle, mounted in a self-alining fixture to be inserted in the breech of the weapon so that.the cross hairs OF the telescope fall on the centerline of the bore. The scope shall have a separate focus for reticle and field and be adjustable from 5 yards to 5000 yards. The reticle shall be centered with respect to the O.D. of the adapter within 0.1 mil. The bore scope shall have a 20x magnification or better.
(e) Parallax shields with a centrally located 1/16-inch maximum diameter holes shall be placed over the eyepiece of the telescopes to eliminate parallax.
(f) A panoramic Telescope M100 and a direct sighting Telescope M99 or M99C.
(g) A calibrated gunners Quadrant M1. A calibrated gunners quandrant shall be defined as a quadrant that has been checked on a sine plate at each of the elevations for which the quadrant will be employed during the test and the deviation in tenths of roils from true elevation recorded in a table which will be furnished with the quadrant. The readings of the quadrant shall be corrected accordingly when checking specified tolerances.

4.3 Preliminary checks and adjustments. - The following preliminary checks and adjustments shall be made:

(a) The mounts shall be inspected for tight mechanical installation and proper pinning.
(b) All moving parts on the mounts shall be checked to assure that no interference exists between the fire control instruments and components in the vehicle between the maximum limits of elevation of the Weapon.

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(c) The vehicle shall be cross leveled so that the Gun will track the plumb line within the thickness (1/16 inch) of the line. This shall be done by using the bore scope (see 4.2 (d)) inserted in the breech and setting the nuzzle cross wires on the center of the plumb line, at zero weapon elevation. The weapon shall be elevated to 30 degrees and above tolerance checked. The vehicle shall also be leveled fore and aft so that at zero weapon elevation the weapon will remain at zero ± .1 mil) when traversed 30 degrees to the right or left of center. The above shall be checked with a calibrated gunners quadrant on the breech pads of the weapon. Cross leveling of the vehicle shall be checked periodically throughout the test.

(d) The gun stop settings shall be checked by the calibrated gunner's quadrant to determine conformance to 3.3.1.

(e) The azimuth feedback sector shall be checked as the cab is traversed to its maximun limits left and right to determine that the sector does not run off the pinion.

(f) The micro-switch circuit shall be checked. to insure that the switch opens when the bell crank opens. This check not required on telescope mount M99A1C.

4.4 Tests for mounts M99 and M99A1C.

4.4.1 Alinement of panoramic telescope. - A panoramic telescope shall be inserted in the mount socket and locked in place. A parallax shield shall be placed over the eyepiece. The cross leveling mechanism shall be adjusted to give least movement or the line of sight of the panoramic telescope from a target point through 360 degree rotation of the azimuth table. At zero elevation, the vertical cross hair of the panoramic telescope shall be placed on the aiming point. The weapon shall be elevated from zero to 1100 mils and checked for conformance to 3.3.2.

4.4.2 Levelness of azimuth table - Without altering the cross leveling adjustments made in 4.4.1, the azimuth table shall be rotated through 360 degrees and checked for conformance to 3.3.3.

4.4.3 Horizontal deflection of panoramic telescope. - A gradual steady pull of 50 pounds shall be applied to the eyepiece of the telescope, tangentially to the right and perpendicular to the axis of the telescope, and then gradually released. While sighting through the telescope, the vertical cross hair shall be laid on the center of the panoramic telescope aiming point. A pull of 50 pounds shall be applied to the eyepiece in the opposite direction and gradually released. The deflection of the vertical cross hair relative to the aiming point shall be checked for conformance to 3.3.4.

4.4.4 Accuracy of elevation counter. - A calibrated gunner's quadrant shall be installed on the breech pads of the weapon and the difference between counter reading and gun elevation checked and noted. The check points shall include minus 55, zero, 225, 425, 1025, and. 1155 mils counter elevation. at each setting, the counter indexes shall be made coincident by laying the Gun from low to high to exclude backlash and true gun elevation shall be read, by the gunner’s quadrant, to determine conformancwe to 3.3.5

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4.4.4.1 Backlash. - The elevation counter indexes shall be made coincident by laying the gun low to high and reading gun elevation by use of the gunner's quadrant. The elevation counter indexes shall again be made coincident by laying the gun from high to low and the gunner's quadrant shall again be read. The difference between the two readings is the backlash at the given elevation and shall be noted for conformance to 3.3.5.1.

4.4.5 Boresight check of panoramic telescope mount. - The weapon shall be aimed on its proper aiming point on the boresight retention and checking target by viewing through the weapon bore (see 4.2 (d)).With the panoramic telescope installed in the mount, position shall be noted to determine conformance to 3.3.6.

4.5 Tests for direct sighting telescope mount M101.

4.5.1 Alinement of telescope. - The direct sighting telescope shall be installed in mount M101 and locked in place. Viewing through the telescope with the parallax shield ins'tiled over the eyepiece, the line of sight shall be set on the plumb line at zero elevation of the weapon. The gun shall be elevated throgh 30 degrees elevation and position noted to determine conformance to 3.3.7.

4.5.2 Boresight check of direct fire telescope. - The weapon shall be placed on its aiming point on the boresight retention and checking target and by viewing through the weapon bore (see 4.2 (d)), location of the line of sight on the target shall-be noted to determine conformance to 3.3.8.

4.6 Boresight retention tests. - After installation on the vehicle and prior to the 10-mile road test specified in Specification MIL-G-L0644 and MIL-H-11964, the cross hairs of the panoramic telescope and direct sighting telescope shall be placed on the center of their respective aiming points shown in Figure 1 after laying the weapon exactly on center. Following the 10-mile road test the vehicle shall be returned to the place from which the original settings were made. Approaching the weapon aiming point from the same direction as before the test, the weapon shall be boresighted at proper aiming point on the target and observation shall be made through the panoramic telescope and direct sighting telescope to determine conformance to 3.3.9.

5. PREPARATION FOR DELIVERY
Not applicable to this specification.

6. NOTES

6.1 Ordering data. - Contracts or orders should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable drawings (see 3.1).

6.2 Telescope. - Taylor Robson No. 112/392 telescope has been found to be satisfactory for the telescope specified in 4.2 (d).

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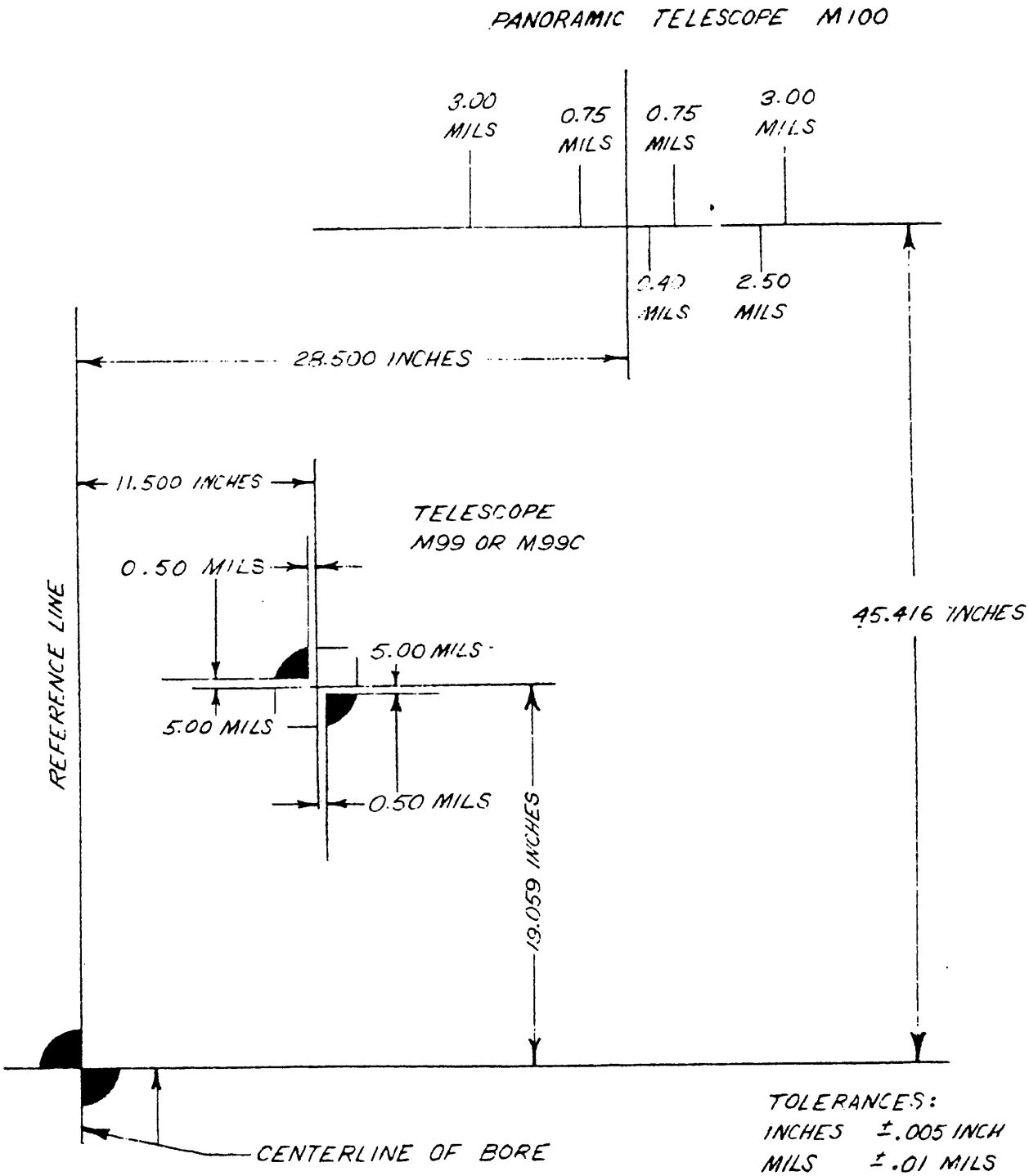

FIGURE 1

BORESIGHT RETETENTION AND CHECKING TARGET FOR SELF-PROPELLED GUN M-53 AND SELF-PROPELLED HOWITZER M-55.

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